TUTORIAL 4

Complex Polynomials and Matrix Operations

4.1 Polynomials over the Complex Numbers

DEFINITION 4.1

The notation $\mathbb{C}[X]$ denotes the polynomials in the variable X with complex coefficients:

$$p(X) = a_0 + a_1 X + \cdots + a_n X^n, \qquad a_j \in \mathbb{C}.$$

Two polynomials are equal when the coefficients of each power agree. Addition adds the coefficients of each power; multiplication by a complex number multiplies every coefficient. Polynomial multiplication expands brackets using $X^j X^k = X^{j+k}$ and collects equal powers. If $a_n \neq 0$, the degree is $\deg p = n$, and a_n is its leading coefficient. A monic polynomial has leading coefficient 1. The zero polynomial has every coefficient zero; we will not assign it a numerical degree here. The phrase *degree at most n* will also allow the zero polynomial.

NOTE

For nonzero polynomials p, q, we have $\deg(pq) = \deg p + \deg q$: the leading coefficient of the product is the product of the two nonzero leading coefficients. In particular, a product of nonzero polynomials is not the zero polynomial.

PROPOSITION 4.1

For $p, d \in \mathbb{C}[X]$ with $d \neq 0$, there are unique polynomials q, r such that

$$p = dq + r, \qquad r = 0 \text{ or } \deg r < \deg d.$$

These are the quotient and remainder when dividing p by d. Long division repeatedly subtracts a multiple of d that cancels the leading term, so the degree decreases. For uniqueness, if $p = dq + r = dq' + r'$, then $d(q - q') = r' - r$. If $q - q' \neq 0$, the left side has degree at least $\deg d$, whereas the right side is zero or has smaller degree. Thus $q = q'$ and then $r = r'$.

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PROPOSITION 4.2

A number $a \in \mathbb{C}$ is a root of p when $p(a) = 0$. We have

$$p(a) = 0 \quad \Longleftrightarrow \quad p(X) = (X - a)q(X)$$

for some polynomial q.

Proof.
On division by $X - a$, the remainder is a constant r, so $p(X) = (X - a)q(X) + r$. Substitution at $X = a$ gives $r = p(a)$, proving both directions.

PROPOSITION 4.3

A nonzero polynomial of degree n has at most n distinct roots.

Proof.
For $n = 0$, the polynomial is a nonzero constant and has no roots. If there is a root a, write $p = (X - a)q$ with $\deg q = n - 1$. Every other root $b \neq a$ satisfies $0 = (b - a)q(b)$, so $q(b) = 0$. Induction gives at most $n - 1$ such roots, or at most n including a.

DEFINITION 4.2

For $p \neq 0$ and an integer $m \geq 1$, a root a has multiplicity m if

$$p(X) = (X - a)^m q(X), \qquad q(a) \neq 0.$$

Thus exactly m factors $X - a$ can be removed. We do not assign a finite root multiplicity to the zero polynomial.

PROPOSITION 4.4

The fundamental theorem of algebra. Every nonconstant complex polynomial has a complex root. Consequently, if $\deg p = n \geq 1$ and its leading coefficient is c, then

$$p(X) = c(X - \lambda_1) \cdots (X - \lambda_n).$$

In this list, each root occurs according to its multiplicity. We will use this theorem without proof.

NOTE

For $p(X) = \sum_{k=0}^{n} a_k X^k$, define $\overline{p}(X) = \sum_{k=0}^{n} \overline{a_k} X^k$ by conjugating the coefficients. Then $\overline{p}(\overline{z}) = \overline{p(z)}$. If p has real coefficients, then $\overline{p} = p$, so the conjugate $\overline{a}$ of every root a is also a root. Multiplicity is preserved as well: from $p = (X - a)^m q$ with $q(a) \neq 0$, coefficient conjugation gives

$$p(X) = (X - \overline{a})^m \overline{q}(X), \qquad \overline{q}(\overline{a}) = \overline{q(a)} \neq 0.$$

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DEFINITION 4.3

The formal derivative of $p(X) = \sum_{k=0}^{n} a_k X^k$ is

$$p'(X) = \sum_{k=1}^{n} k a_k X^{k-1}.$$

This is an operation on coefficients. The product rule $(fg)' = f'g + fg'$ holds: for $j, k \geq 1$, it is the identity $(X^j X^k)' = (j+k)X^{j+k-1} = jX^{j-1}X^k + X^j kX^{k-1}$; if one exponent is zero, the rule follows because a constant has derivative zero. Expansion and addition give the rule for all polynomials.

PROPOSITION 4.5

For $p \neq 0$, the root a has multiplicity at least 2 if and only if $p(a) = p'(a) = 0$.

Proof.
If $p(a) = 0$, write $p = (X - a)q$. The product rule gives

$$p' = q + (X - a)q', \qquad p'(a) = q(a).$$

Thus $p'(a) = 0$ is equivalent to q also being divisible by $X - a$, which means that p is divisible by $(X - a)^2$.

EXERCISE 4.1

Please factor $P(X) = (X - 1)^4 + 4$ into linear factors over $\mathbb{C}$, and please find all its roots and their multiplicities.

WORKED SOLUTION

Set $Y = X - 1$. A difference of squares gives

$$Y^4 + 4 = (Y^2 + 2)^2 - (2Y)^2 = (Y^2 - 2Y + 2)(Y^2 + 2Y + 2).$$

Returning to X,

$$P(X) = (X^2 - 4X + 5)(X^2 + 1) = \big((X - 2)^2 + 1\big)(X^2 + 1).$$

Therefore

$$P(X) = (X - i)(X + i)(X - 2 - i)(X - 2 + i).$$

The roots are $i, -i, 2 + i, 2 - i$. They are distinct, so each has multiplicity 1.

That does it. Victory!

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EXERCISE 4.2

Please find a real-coefficient polynomial of the smallest possible degree for which $1+2i$ is a root of multiplicity 2 and $p(0)=50$. Please determine whether such a polynomial of smallest degree is unique.

WORKED SOLUTION

The conjugate $1-2i$ must also be a root of multiplicity 2. The two roots are distinct, so the degree is at least 4. In degree 4, the polynomial must be

$$p(X) = c(X-1-2i)^2(X-1+2i)^2 = c(X^2-2X+5)^2.$$

The condition $p(0)=50$ gives $25c=50$, hence $c=2$. Thus

$$p(X) = 2(X^2-2X+5)^2 = 2X^4-8X^3+28X^2-40X+50.$$

This polynomial satisfies all conditions. The root $1+2i$ has multiplicity exactly 2 because $1+2i \neq 1-2i$. The value at 0 determines c, so the polynomial of smallest degree is unique.

All wrapped up. Victory!

EXERCISE 4.3

Let $n \geq 0$ be an integer. Please find the remainder when X^{4n+3} is divided by X^2-2X+2.

WORKED SOLUTION

The divisor has roots $\alpha = 1+i$ and $\beta = 1-i$. The remainder has the form $r(X) = aX+b$. Substituting the roots gives

$$a\alpha + b = \alpha^{4n+3}, \qquad a\beta + b = \beta^{4n+3}.$$

We compute

$$(1\pm i)^2 = \pm 2i, \qquad (1\pm i)^4 = -4, \qquad (1\pm i)^3 = -2\pm 2i.$$

Hence

$$a(1+i)+b = (-4)^n(-2+2i), \qquad a(1-i)+b = (-4)^n(-2-2i).$$

Subtracting gives $2ia = 4i(-4)^n$, so $a = 2(-4)^n$. Substitution in either equation gives $a+b = -2(-4)^n$, hence $b = -4(-4)^n$. The remainder is

$$r(X) = 2(-4)^n(X-2).$$

It has degree less than 2 and the required values at both roots. Uniqueness of quotient and remainder gives the result.

We've reached the finish. Victory!

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EXERCISE 4.4

Please find all $p \in \mathbb{C}[X]$ of degree at most 3 satisfying

$$p(i) = p'(i) = 0, \qquad p(-i) = 2, \qquad p'(-i) = 0.$$

WORKED SOLUTION

The conditions at i imply $p = (X-i)^2(aX+b)$ for some $a, b \in \mathbb{C}$. The value $p(-i) = 2$ gives

$$(-2i)^2(-ia+b) = 2 \quad \Longrightarrow \quad b - ia = -\frac{1}{2}.$$

Differentiation gives

$$p'(X) = 2(X-i)(aX+b) + a(X-i)^2.$$

Thus $p'(-i) = 0$ becomes

$$-4i(b-ia) - 4a = 0.$$

Substituting $b - ia = -1/2$ gives $2i - 4a = 0$, so $a = i/2$. Now $b - i(i/2) = -1/2$, hence $b = -1$. The unique polynomial is

$$p(X) = (X-i)^2\left(\frac{i}{2}X - 1\right) = 1 + \frac{i}{2}(X^3 + 3X).$$

Indeed, $p(i) = 0$, $p(-i) = 2$, and $p'(X) = \frac{3i}{2}(X^2+1)$ vanishes at both points. Uniqueness also follows directly: the difference of two solutions would be divisible by $(X-i)^2(X+i)^2$, a degree-4 polynomial, although its degree is at most 3. The difference must therefore be zero.

Yes! Victory!

EXERCISE 4.5

Let $p \in \mathbb{C}[X]$ have degree $n \geq 1$. Please prove that, except for at most $n-1$ values of $w \in \mathbb{C}$, the equation $p(z) = w$ has exactly n distinct solutions. Please find the exceptional values for $p(X) = X^3 - 3X$.

WORKED SOLUTION

The derivative p' has degree $n-1$, so it has at most $n-1$ distinct roots. Let

$$S = \{p(c) : c \in \mathbb{C},\ p'(c) = 0\}$$

be the set of possible exceptional values. Its number of elements is at most the number of roots of p', so $|S| \leq n-1$. If $w \notin S$, the polynomial $p - w$ has no repeated root: such a root c would satisfy both $p(c) = w$ and $p'(c) = 0$, contradicting the choice of w. The fundamental theorem of algebra gives n roots counted with multiplicity. They are all simple, hence n distinct solutions. Conversely, if $w \in S$, the polynomial $p - w$ has a repeated root and therefore fewer than n distinct roots. For the given example,

$$p'(X) = 3(X^2-1), \qquad p(1) = -2, \qquad p(-1) = 2.$$

The exceptional values are exactly $-2, 2$. Indeed,

$$p(X) - 2 = (X-2)(X+1)^2, \qquad p(X) + 2 = (X+2)(X-1)^2.$$

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EXERCISE 4.5 CONTINUED

Each exceptional value has two distinct solutions; every other value has three.

V-I-C-T-O-R-Y!

EXERCISE 4.6

Let $p(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0$ be monic, with $n \geq 1$, and set $a_n = 1$. Suppose all its roots lie on the unit circle. Please prove that

$$|a_0| = 1, \qquad a_k = a_0\overline{a_{n-k}} \quad (0 \leq k \leq n).$$

Please determine whether these coefficient conditions guarantee that all roots lie on the unit circle.

WORKED SOLUTION

Write $p(X) = (X-\lambda_1)\cdots(X-\lambda_n)$, listing each root according to its multiplicity, with $|\lambda_j| = 1$. The constant coefficient is

$$a_0 = (-1)^n\lambda_1\cdots\lambda_n,$$

so $|a_0| = 1$. Define a new polynomial by reversing and conjugating the coefficients:

$$p^{\#}(X) = \sum_{k=0}^{n}\overline{a_{n-k}}X^k = (1-\overline{\lambda_1}X)\cdots(1-\overline{\lambda_n}X).$$

Since $\overline{\lambda_j}\lambda_j = 1$, each factor satisfies

$$1-\overline{\lambda_j}X = -\overline{\lambda_j}(X-\lambda_j).$$

Consequently,

$$p^{\#}(X) = (-1)^n\overline{\lambda_1\cdots\lambda_n}\,p(X) = \overline{a_0}\,p(X).$$

Comparison of coefficients of X^k gives $\overline{a_{n-k}} = \overline{a_0}a_k$. Multiplying by a_0 and using $a_0\overline{a_0} = 1$ gives the required identity. The conditions are not sufficient. For example,

$$p(X) = X^2 - \frac{5}{2}X + 1 = (X-2)\left(X-\frac{1}{2}\right)$$

satisfies them, but neither root lies on the unit circle.

🏆 **Victory!**

NOTATION AND CONVENTIONS

Let F be a field. The set $M_{m\times n}(F)$ consists of matrices with m rows, n columns, and entries in F; $M_n(F)$ consists of square matrices of order n. The entry A_{ij} is in row i and column j. The column $e_j \in F^n$ has entry 1 in position j and zero elsewhere. For a square matrix, the trace is $\operatorname{tr} A = \sum_{i=1}^{n} A_{ii}$. The identity matrix I_n has diagonal entries 1 and all other entries 0. The zero matrix has only zero entries.

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4.2 Matrix Arithmetic

Throughout this section, let

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 2 \\ -1 & 5 \end{pmatrix}.$$

4.2.1 Scalar Multiplication

Scalar multiplication is defined for every matrix. Multiply **every** entry by the scalar: $(\lambda A)_{ij} = \lambda(A)_{ij}$.

EXAMPLE 4.1

Compute

$$-3A = \begin{pmatrix} -3 \cdot 2 & (-3)(-1) \\ -3 \cdot 3 & -3 \cdot 4 \end{pmatrix} = \begin{pmatrix} -6 & 3 \\ -9 & -12 \end{pmatrix}.$$

4.2.2 Matrix Addition and Subtraction

These are defined only for two matrices of **exactly the same shape**. Add or subtract corresponding entries: $(A \pm B)_{ij} = (A)_{ij} \pm (B)_{ij}$.

EXAMPLE 4.2

Compute

$$A + B = \begin{pmatrix} 2+0 & -1+2 \\ 3-1 & 4+5 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & 9 \end{pmatrix}.$$

4.2.3 Matrix Multiplication

A product is defined only when the number of columns of the first (left) matrix equals the number of rows of the second (right) matrix. If A is $k \times m$ and B is $m \times n$, then AB is $k \times n$ and is defined by

$$(AB)_{ij} = \sum_{r=1}^{m} a_{ir} b_{rj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{im}b_{mj}.$$

This is the entry in row i, column j of the product. Follow row i of A and column j of B, each containing m entries: multiply matching entries and add. **Note:** the existence of AB does **not** ensure the existence of BA. With the dimensions above, the latter also requires $n = k$.

EXERCISE 4.7

Please compute AB and BA for the matrices defined above.

WORKED SOLUTION

$$AB = \begin{pmatrix} 2 \cdot 0 + (-1)(-1) & 2 \cdot 2 + (-1)5 \\ 3 \cdot 0 + 4(-1) & 3 \cdot 2 + 4 \cdot 5 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -4 & 26 \end{pmatrix},$$

$$BA = \begin{pmatrix} 0 \cdot 2 + 2 \cdot 3 & 0(-1) + 2 \cdot 4 \\ (-1)2 + 5 \cdot 3 & (-1)(-1) + 5 \cdot 4 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 13 & 21 \end{pmatrix}.$$

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EXERCISE 4.7 CONTINUED

Each entry is the sum of the matching row-column products. Reversing the order changes the result.

And we're done. Victory!

NOTE

Important conclusion: matrix multiplication is not commutative. The equality $AB = BA$ need not hold. In the example, $AB \neq BA$ even though both products are defined.

EXERCISE 4.8

Let A be a $k \times m$ matrix with a zero row, and let B be $m \times n$. Please prove that the $k \times n$ matrix AB has a zero row.

WORKED SOLUTION

Let i be the index of a zero row of A, where $1 \leq i \leq k$. Thus $(A)_{ir} = 0$ for every $1 \leq r \leq m$. In row i of AB, for every $1 \leq j \leq n$,

$$(AB)_{ij} = \sum_{r=1}^{m} (A)_{ir}(B)_{rj} = \sum_{r=1}^{m} 0 = 0.$$

All entries in row i of AB are zero. **The zero matrix** is a matrix of any specified shape whose entries are all zero.

You know what that means? V-I-C-T-O-R-Y!

4.3 Diagonal Matrices

A **diagonal matrix** is a **square** matrix whose off-diagonal entries are all zero. Write

$$D = \operatorname{diag}(d_1, \ldots, d_n) = \begin{pmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{pmatrix}.$$

Equivalently,

$$[D]_{ij} = \begin{cases} 0, & i \neq j, \\ d_i, & i = j. \end{cases}$$

A **scalar matrix** is a diagonal matrix with equal diagonal entries. The **identity matrix** is a diagonal matrix with all diagonal entries equal to 1, hence is a scalar matrix. A scalar matrix can therefore be written aI, where a is its common diagonal entry. Write I_n for the $n \times n$ identity, or simply I when its size is clear. For every $n \times m$ matrix A, $AI_m = I_n A = A$.

EXAMPLE 4.3

$$I_2 A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}, \qquad AI_2 = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}.$$

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EXERCISE 4.9

Let A, B be diagonal matrices of order n. Please prove that:

1. $A + B$ is diagonal.
2. AB is diagonal and $[AB]_{ii} = a_{ii}b_{ii}$ for every $1 \leq i \leq n$.
3. A and B **commute**: $AB = BA$.

WORKED SOLUTION

1. For $i \neq j$, $(A + B)_{ij} = a_{ij} + b_{ij} = 0 + 0 = 0$. Thus $A + B$ is diagonal, with diagonal entries $(A + B)_{ii} = a_{ii} + b_{ii}$.
2. By the definition of multiplication,

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}.$$

   In row i of A, only a_{ii} can be nonzero, so the sum reduces to $a_{ii}b_{ij}$. If $i \neq j$, then $b_{ij} = 0$, giving $(AB)_{ij} = 0$. If $i = j$, then $(AB)_{ii} = a_{ii}b_{ii}$. Hence AB is diagonal.
3. Similarly BA is diagonal with diagonal entries $b_{ii}a_{ii}$. Scalars in a field commute, so

$$a_{ii}b_{ii} = b_{ii}a_{ii} \qquad (1 \leq i \leq n).$$

   All entries of AB and BA agree; therefore $AB = BA$.

Another one down. Victory!

NOTE

Check dimensions before computing. The product $A_{k\times m}B_{m\times n}$ has k rows and n columns. Equality of matrices requires both equal dimensions and equal corresponding entries.

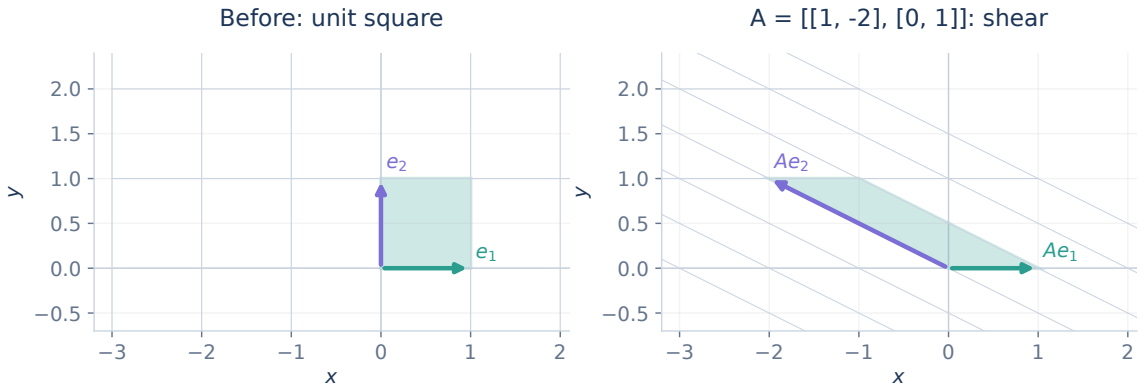


Figure 4.1. A linear map takes a straight grid to a straight grid. The columns of the matrix describe the images of the basis vectors.

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4.4 Transpose

- The transpose is defined for every matrix. The **transpose** A^T satisfies $(A^T)_{ij} = (A)_{ji}$.
- Rows become columns and columns become rows.
- If $A \in \mathbb{R}^{m\times n}$, then $A^T \in \mathbb{R}^{n\times m}$: the numbers of rows and columns are interchanged.

EXAMPLE 4.4

$$C = \begin{pmatrix} 2 & -3 \\ 0 & 7 \\ -4 & 1 \end{pmatrix} \in \mathbb{R}^{3\times 2}, \qquad C^T = \begin{pmatrix} 2 & 0 & -4 \\ -3 & 7 & 1 \end{pmatrix} \in \mathbb{R}^{2\times 3}.$$

The rows of C are the columns of C^T.

4.5 Symmetry and Skew-Symmetry

- A **square** matrix A is **symmetric** if $A^T = A$.
- A **square** matrix A is **skew-symmetric** if $A^T = -A$.
- Every diagonal matrix is symmetric; in particular, every scalar matrix and the identity matrix are symmetric.

The square condition matters: $A \in \mathbb{F}^{m\times n}$ and $A^T \in \mathbb{F}^{n\times m}$ have the same shape only when $n = m$. Intuitively, symmetry means reflection across the main diagonal; skew-symmetry means the same reflection followed by a change of sign.

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EXERCISE 4.10

Please classify each matrix as symmetric, skew-symmetric, or neither.

$$A = \begin{pmatrix} 2 & -4 \\ -4 & 7 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 2 & 1 \\ -5 & 0 \end{pmatrix}.$$

WORKED SOLUTION

Computing transposes gives

$$A^T = \begin{pmatrix} 2 & -4 \\ -4 & 7 \end{pmatrix} = A, \qquad B^T = \begin{pmatrix} 0 & -3 \\ 3 & 0 \end{pmatrix} = -B,$$

$$C^T = \begin{pmatrix} 2 & -5 \\ 1 & 0 \end{pmatrix} \neq C, \qquad C^T \neq -C.$$

Thus A is symmetric, B is skew-symmetric, and C is neither.

We've got it. Victory!

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EXERCISE 4.11

Please prove or disprove.

1. If A, B are symmetric, then AB is symmetric.
2. If A, B are both nonsymmetric, then AB is nonsymmetric.

WORKED SOLUTION

1. $(AB)^T = B^T A^T = BA$, so symmetry of both factors does not ensure symmetry of the product. For example,

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & -1 \\ -1 & 3 \end{pmatrix}, \quad AB = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}.$$

   Both factors are symmetric, but the product is not.

2. Take $A = B = \begin{pmatrix} 0 & 2 \\ -3 & 0 \end{pmatrix}$. Neither is symmetric, but

$$AB = \begin{pmatrix} 0 \cdot 0 + 2(-3) & 0 \cdot 2 + 2 \cdot 0 \\ (-3)0 + 0(-3) & (-3)2 + 0 \cdot 0 \end{pmatrix} = -6I_2.$$

   The product is symmetric. Thus both assertions are false.

That does it. Victory!

EXERCISE 4.12

Let $A \in M_n(\mathbb{F})$, where $\text{char}(\mathbb{F}) \neq 2$. Please prove that:

1. $A + A^T$ is symmetric.
2. $A - A^T$ is skew-symmetric.
3. A can be written as the sum of a symmetric matrix and a skew-symmetric matrix.

WORKED SOLUTION

1. $(A + A^T)^T = A^T + (A^T)^T = A^T + A = A + A^T$.
2. $(A - A^T)^T = A^T - A = -(A - A^T)$, proving skew-symmetry.
3. Add the matrices from the preceding parts:

$$2A = (A + A^T) + (A - A^T),$$
$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T).$$

   As shown above, the first summand is symmetric and the second is skew-symmetric.

All wrapped up. Victory!

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Additional practice

EXERCISE 4.13 FOUNDATION

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 0 & 3 \end{pmatrix}.$$

Please determine the dimensions of AB, BA and please compute both. Please use $e_2 \in \mathbb{R}^2$ to verify $(AB)e_2 = A(Be_2)$. Can $AB = BA$?

WORKED SOLUTION

The dimensions are 2×2 and 3×3, respectively.

$$AB = \begin{pmatrix} 1 \cdot 2 + 0(-1) - 1 \cdot 0 & 1 \cdot 1 + 0 \cdot 0 - 1 \cdot 3 \\ 0 \cdot 2 + 2(-1) + 1 \cdot 0 & 0 \cdot 1 + 2 \cdot 0 + 1 \cdot 3 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ -2 & 3 \end{pmatrix},$$

$$BA = \begin{pmatrix} 2 \cdot 1 + 1 \cdot 0 & 2 \cdot 0 + 1 \cdot 2 & 2(-1) + 1 \cdot 1 \\ (-1)1 + 0 \cdot 0 & (-1)0 + 0 \cdot 2 & (-1)(-1) + 0 \cdot 1 \\ 0 \cdot 1 + 3 \cdot 0 & 0 \cdot 0 + 3 \cdot 2 & 0(-1) + 3 \cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 & -1 \\ -1 & 0 & 1 \\ 0 & 6 & 3 \end{pmatrix}.$$

Here $Be_2 = (1, 0, 3)^T$, so $A(Be_2) = (-2, 3)^T$. Also $(AB)e_2 = (-2, 3)^T$, the second column of AB. The dimensions differ, so AB and BA cannot be equal.

We've reached the finish. Victory!

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EXERCISE 4.14 CHALLENGE

Let $n \geq 2$ and $A \in M_n(\mathbb{F})$ over an arbitrary field. Please prove:

1. A commutes with every diagonal matrix if and only if A is diagonal.
2. A commutes with every matrix in $M_n(\mathbb{F})$ if and only if $A = \lambda I_n$ for some $\lambda \in \mathbb{F}$.

You may use the matrix units E_{ij}, whose entry (i, j) is 1 and all other entries are zero.

WORKED SOLUTION

Suppose A commutes with every diagonal matrix. In particular, $AE_{kk} = E_{kk}A$ for every k. In entry (i, k) with $i \neq k$, the left side has value a_{ik} and the right side has value zero. Hence $a_{ik} = 0$ for every $i \neq k$, so A is diagonal. Conversely, two diagonal matrices commute because their product is diagonal and its diagonal entries are products of commuting scalars.
If A commutes with every matrix, part 1 gives $A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. For $i \neq j$,

$$AE_{ij} = \lambda_i E_{ij}, \qquad E_{ij}A = \lambda_j E_{ij}.$$

Comparing entry (i, j) gives $\lambda_i = \lambda_j$. All diagonal entries are therefore equal, so $A = \lambda I_n$. Conversely, for every B, $(\lambda I_n)B = \lambda B = B(\lambda I_n)$. The proof uses only the scalars $0, 1$ and field multiplication, so it applies to finite fields as well.

Yes! Victory!

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